Calculus I Lesson 19

Quotient Rule for Derivative: $D_x\left(\frac{f(x)}{g(x)}\right)=\frac{g(x)\cdot f'(x)-f(x)\cdot g'(x)}{[g(x)]^2}$

1) $h(x)=\frac{x^3+4}{x-7}$

a) $D_x\left(x^3+4\right)=$ ___?___

b) $D_x\left(x-7\right)=$ ___?___

c) $h'(x)=$ ___?___

2) $h(x) = \dfrac{x^2}{\sin x}$

a) $D_x\left(x^2\right) = \underline{\quad ? \quad}$

b) $D_x(\sin x) = \underline{\quad ? \quad}$

c) $h'(x) = \underline{\quad ? \quad}$

3) $h(x) = \dfrac{x^2 + 5x}{2x^2 + 5x - 5}$

a) $D_x\left(x^2 + 5x\right) = \underline{\quad ? \quad}$

b) $D_x\left(2x^2 + 5x - 5\right) = \underline{\quad ? \quad}$

c) $h'(x) = \underline{\quad ? \quad}$

4) $h(x) = \dfrac{\cos x}{5 + \sin x}$

a) $D_x(\cos x) = \underline{\quad ? \quad}$

b) $D_x(5 + \sin x) = \underline{\quad ? \quad}$

c) $h'(x) = \underline{\quad ? \quad}$

5) $h(x) = \dfrac{x^2}{\sin x}$

a) $D_x\left(x^2\right) = \underline{\quad ? \quad}$

b) $D_x(\sin x) = \underline{\quad ? \quad}$

c) $h'(x) = \underline{\quad ? \quad}$

6) $h(x) = \dfrac{x^{-2}}{2x^2 + \sin x}$

a) $D_x\left(x^{-2}\right) = \underline{\quad ? \quad}$

b) $D_x\left(2x^2 + \sin x\right) = \underline{\quad ? \quad}$

c) $h'(x) = \underline{\quad ? \quad}$

7) $h(x)=\dfrac{x^{2/3}+1}{5x^2+\sin x}$

a) $D_x\left(x^{2/3}+1\right)=$ ___?___

b) $D_x\left(5x^2+\sin x\right)=$ ___?___

c) $h'(x)=$ ___?___

8) $h(x) = \dfrac{x^{-3/4} + 7x}{x^3 + \cos x}$

a) $D_x\left(x^{-3/4} + 7x\right) = \underline{\quad ? \quad}$

b) $D_x\left(x^3 + \cos x\right) = \underline{\quad ? \quad}$

c) $h'(x) = \underline{\quad ? \quad}$

9) $h(x) = \dfrac{x^{-5/4} + 2x^3}{x^{1/2} + \cos x}$

a) $D_x\left(x^{-5/4} + 2x^3\right) = \underline{\quad ? \quad}$

b) $D_x\left(x^{1/2} + \cos x\right) = \underline{\quad ? \quad}$

c) $h'(x) = \underline{\quad ? \quad}$